Session 10: Optimization Methods for Mission Design

Task 1: Reading and Notes

1. Standard Form Formulation

An optimization problem in mission design is formulated mathematically in standard form as:

$$\min_x J(x) \quad \text{subject to} \quad g(x) \leq 0, \quad h(x) = 0$$

- **Objective Function ($J(x)$):** The scalar quantity to be minimized or maximized (e.g., total Δv, satellite count, propellant mass).
- **Design Variables (x):** The adjustable parameters (e.g., burn magnitudes, impulses, orbital inclinations, semi-major axis, epoch)
- **Constraints:**
  - **Equality constraints ($h(x) = 0$):** Mandatory conditions that must be strictly satisfied (e.g., reaching a specific orbital target).
  - **Inequality constraints ($g(x) \leq 0$):** Boundary limits that must not be violated (e.g., maximum g-load, delta-v budget limits).

2. Concept of the Pareto Front:

- **Pareto Front Definition:** In multi-objective optimization where objectives conflict (e.g., cost vs. performance), no single global optimum exists. The **Pareto front** represents the set of all non-dominated design points.
- **Dominance Condition:** A solution x_1 dominates x_2 if x_1 is at least as good as x_2 in all objectives and strictly better in at least one ($J_i(x_1) \leq J_i(x_2)\ \forall i$).
- **Why a point off the Pareto front is never chosen:** Any point lying off (above or to the right of) the Pareto front is sub-optimal/wasteful. At least one objective can be improved for free without degrading any other objective

Task 2: One-Variable Optimum — Splitting a 51.6° Plane Change

1. Problem Setup:

A LEO-to-GEO transfer starting from an inclination of $i = 51.6^\circ$ (typical for ISS launches). The total inclination change of 51.6° is split between the perigee burn (fraction f) and the apogee burn (fraction $1 - f$).

- **LEO Orbit:** $\boldsymbol{r_1 = R_e + 185}$ km $= \mathbf{6563.137}$ km
- **GEO Orbit:** $\boldsymbol{r_2 = 42164}$ km
- **Transfer Orbit Semi-Major Axis:** $a_t = \frac{r_1 + r_2}{2} = 24363.5685$ km
- **Total Plane Change:** $\Delta i = 51.6^\circ$

2. Velocity Calculations

$$\mu = 398600.4418 \text{ km}^3/\text{s}^2$$

- $v_{\text{LEO}} = \sqrt{\frac{\mu}{r_1}} \approx 7.793$ km/s
- $v_p = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_t}\right)} \approx 10.252$ km/s
- $v_a = \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_t}\right)} \approx 1.596$ km/s
- $v_{\text{GEO}} = \sqrt{\frac{\mu}{r_2}} \approx 3.075$ km/s

3. Combined Delta-v Equation:

$$\Delta v_1(f) = \sqrt{v_{\text{LEO}}^2 + v_p^2 - 2v_{\text{LEO}}v_p \cos(f \cdot \Delta i)}$$

$$\Delta v_2(f) = \sqrt{v_a^2 + v_{\text{GEO}}^2 - 2v_a v_{\text{GEO}} \cos\big((1-f) \cdot \Delta i\big)}$$

$$\Delta v_{\text{total}}(f) = \Delta v_1(f) + \Delta v_2(f)$$

4. Calculated Results:

- **All-at-apogee transfer ($f = 0$):**

$$\begin{aligned}\Delta v_{\text{total}}(0) = {} & (10.252 - 7.793) \\ & + \sqrt{1.596^2 + 3.075^2 - 2(1.596)(3.075)\cos(51.6°)} \\ & \approx 4.889 \text{ km/s}\end{aligned}$$

- **Optimum — Minimizing Total Delta-v (f^*)**
  - Solving $\frac{d(\Delta v_{\text{total}})}{df} = 0$ numerically
  - Optimal Fraction at Perigee (f^*):

$$f^* \approx 0.0541 \Longrightarrow \mathbf{5.4\%}$$

  - **Inclination Split:**
    - At Perigee: $\Delta i_1 = 0.0541 \times 51.6° \approx 2.79°$
    - At Apogee: $\Delta i_2 = 51.6° - 2.79° = 48.81°$
  - **Component Delta-v at Optimum:**

- $\Delta v_1(f^*) = \sqrt{7.793^2 + 10.252^2 - 2(7.793)(10.252)\cos(2.79°)} \approx 2.497$ km/s
- $\Delta v_2(f^*) = \sqrt{1.596^2 + 3.075^2 - 2(1.596)(3.075)\cos(48.81°)} \approx 2.314$ km/s

- **Minimum Total Delta-v:**
  - $\Delta v_{\text{total}}(f^*) = 2.497 + 2.314 = \mathbf{4.811}$ km/s

- **Δv Savings:**

$$\Delta v_{\text{saving}} = \Delta v_{\text{total}}(0) - \Delta v_{\text{total}}(f^*) = 4.889 - 4.811 = 0.078\ km/s$$

Task 3: Pareto Reasoning

Based on the constellation trade study in Figure 2:

- **Selection A (Mean Revisit < 8 minutes at lowest satellite count):**
  - **Selected Design:** The non-dominated constellation with **36 satellites at an altitude of** 1500 km.
  - **Justification:** Examining the Pareto front (dashed red line) in Figure 2, this point achieves a mean revisit time of ≈ 6.8 minutes (satisfying the < 8 min constraint) using the absolute minimum number of satellites (36 spacecraft) on the front.
- **Selection B (Priority: Fewest satellites achieving > 90% coverage):**
  - **Selected Design:** Higher-altitude configurations (e.g., 1500 km) with a reduced constellation count.
  - **Justification:** Satellite coverage footprint scales directly with orbital altitude. Choosing a higher altitude (1500 km) widens each satellite's footprint, allowing the total constellation size to be minimized while still satisfying broad global coverage (>90%).

Task 4: Method Matching

1. **Earth-to-Mars Launch Window:**
   - **Suitable Method: Grid Search / Parametric Sweep (Porkchop Plot).**

- **Reasoning:** The 2-D design space across departure and arrival dates contains multiple localized energy minima. Evaluating a regular parametric grid ensures finding the global minimum launch window without getting stuck in local valleys.

2. **Low-Thrust Spiral Trajectory:**
   - **Suitable Method: Direct Transcription / Collocation.**
   - **Reasoning:** Low-thrust maneuvers involve continuous control inputs. Discretizing the trajectory states and controls transforms the continuous optimal control problem into a non-linear programming (NLP) system solved efficiently via SQP solvers.
3. **Sizing a Walker Constellation:**
   - **Suitable Method: Multi-Objective Global Optimization (e.g., Genetic Algorithm / NSGA-II).**
   - **Reasoning:** Constellation trade spaces combine discrete integer variables (planes, satellite count, phasing) with conflicting trade-off metrics (cost vs. revisit time).
4. **Gravity-Assist Tour of Outer Planets:**
   - **Suitable Method: Global / Heuristic Search.**
   - **Reasoning:** Multi-flyby planetary tours produce highly non-linear, rugged objective functions with deep local minima where gradient methods reliably fail.

Task 5: Code Verification

```
import numpy as np
from scipy.optimize import minimize_scalar


# Constants
mu = 398600.4418      # Earth gravitational parameter (km^3/s^2)
Re = 6378.137         # Earth equatorial radius (km)


# Orbit Parameters
```

```
r1 = Re + 185.0       # LEO radius (km)
r2 = 42164.0          # GEO radius (km)
di = np.radians(51.6) # Total plane change angle (rad)

# Orbital Velocities
a_t = (r1 + r2) / 2.0
vleo = np.sqrt(mu / r1)
vp = np.sqrt(mu * (2.0 / r1 - 1.0 / a_t))
va = np.sqrt(mu * (2.0 / r2 - 1.0 / a_t))
vgeo = np.sqrt(mu / r2)

# Objective Function: Total Delta-v
def total_dv(f):
    # f = fraction of plane change performed at perigee
    dv1 = np.sqrt(vleo**2 + vp**2 - 2.0 * vleo * vp * np.cos(f * di))
    dv2 = np.sqrt(va**2 + vgeo**2 - 2.0 * va * vgeo * np.cos((1.0 - f) * di))
    return dv1 + dv2

# Optimization Execution
res = minimize_scalar(total_dv, bounds=(0, 1), method='bounded')

# Baseline calculation: 100% plane change at apogee (f = 0)
dv_apogee = total_dv(0.0)

# Print Detailed Results
print("--- TASK 2 NUMERICAL CONFIRMATION ---")
print(f"Optimal fraction at perigee: {res.x * 100:.2f}%")
print(f"Minimum total delta-v:      {res.fun:.4f} km/s")
print(f"All-at-apogee delta-v:      {dv_apogee:.4f} km/s")
print(f"Delta-v savings:           {(dv_apogee - res.fun) * 1000.0:.2f} m/s")
```